MIL-S-50087C
17 April 1968
SUPERSEDING
MIL-S-50087B
17 March 1964

MILITARY SPECIFICATION

SEPARATOR, DUST

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a small cyclone dust separator.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

DL5-19-967	-	Separator, Dust.
DL5-19-5381	-	Separator, Dust (Alternate Design).
DL136-14-272	-	Tester, Air Flow, Filter Unit, Q96

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procurement activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Freight Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Materials. All materials cited on Drawing DL5-19-967 or DL5-19-5381 shall conform to the specifications listed therein or to the specific characteristics set forth on the drawings.

3.2 Manufacture and assembly. The separator shall be manufactured and assembled as specified on Drawing D5-19-967 or D5-19-5381.

3.3 Performance. The separator shall have a pressure drop of less than 1.0 inch of water with a measured air input of 12 cubic feet per minute (cfm) and a leakage of 1.5 ± 0.1 cfm with an air inlet pressure of 12 inches of water when tested as specified in 4.3.4.1.

3.4 Preproduction. Prior to the start of regular production, a preproduction sample of dust separators shall be produced in accordance with this specification for examination and test (see 4.2).

3.5 Workmanship. The dust separator shall be free from damage (dented, bent, burred, or cracked metal parts, or chipped finish) and foreign matter (dirt, oil or viscous material).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

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4.2 Preproduction.

4.2.1 Sample. A preproduction sample of 20 dust separators shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.2.2 Inspection.

4.2.2.1 For examination. Each dust separator in the preproduction sample shall be examined for all requirements of the applicable drawings and this specification.

4.2.2.2 For test. Each dust separator in the preproduction sample shall be tested for the requirements of this specification in accordance with 4.3.4.

4.3 Regular production.

4.3.1 Lotting. A lot shall consist of the dust separators produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

4.3.2 Sampling for examination and test. Sampling for examination and test shall be conducted in accordance with MIL-STD-105.

4.3.3 Inspection.

4.3.3.1 For examination and test. Sample dust separators shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

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4.3.3.2 Classification of defects.

(a) Separator, dust (Drawing D5-19-967).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 0.1 percent defective	
101	Performance	4.3.4.1
	AQL 1.0 percent defective	
102	Component missing or incorrectly assembled	
103	Braze incomplete	
104	Hole location 0.843 or 0.125 incorrect	
105	Lug projection not removed (as applicable)	
106	Tube projection incorrect	
107	Tube (cone end) unpainted	
108	Protective finish missing	
109	Hole location incorrect (C5-19-974)	
110	Thread incorrect (C5-19-974)	
111	Workmanship (see 3.5)	

(b) Separator, dust (alternate design Drawing D5-19-5381).

Categories:	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 0.1 percent defective	
101	Performance	4.3.4.1
	AQL 1.0 percent defective	
102	Component missing or incorrectly assembled	
103	Workmanship (see 3.5)	

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(c) Body, dust separator (alternate design Drawing D5-19-3095).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Location of .172 - .179 dia. hole incorrect	
102	Thread (Tap 12 - 24 UNC - 2B thru 3 places) incorrect	
103	Protective finish missing	
104	Flash not removed	
105	Overall dimensions incorrect	
106	Workmanship (see 3.5)	

(d) Plate, core, dust separator (alternate design Drawing D5-19-3096).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 0.1 percent defective	
101	0.036 dimension incorrect (2 places)	
102	0.1330 diameter hole incorrect (2 places)	
	AQL 1.0 percent defective	
103	Hole sizes incorrect	
104	Location of holes incorrect	
105	Overall dimensions incorrect	
106	Flash not removed	
107	Protective finish missing	
108	Workmanship (see 3.5)	

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4.3.4 Test.

4.3.4.1 Performance test. Performance shall be determined with Q96 filter unit airflow tester (DL 136-14-272).

5. PREPARATION FOR DELIVERY

5.1 Packaging, level C. The dust separator shall be packaged to provide adequate protection against corrosion, deterioration, and damage from the supply source to the first receiving activity for immediate use or for controlled humidity storage.

5.2 Packing, level C. Dust separators, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

5.4 Repair part. When this item is procured for storage and issue as a repair part, preservation, packaging, packing and marking shall be as specified on the repair part packaging data sheet which is identified by the Federal stock number for the repair part.

6. NOTES

6.1 Intended use. The dust separator is intended for use as a component of low airflow collective protection equipment.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

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6.3 International interest. Certain provisions of this specification are the subject of international standardization agreement ABCA-ARMY-STD 93 and ABCA-ARMY-STD-154. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through internation standardization channels including departmental standardization offices, if required.

Custodians:

Army - MU

Air Force - 84

Review activities

Navy - MC, YD

User activities:

Navy - SH

International interest (see section 6)

Preparing activity:

Army - MU(EA)

Project Number 4240-0264

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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